

PULCOSIN ORAL SOLUTION

250mg/ml

Ingredient(s):

Each ml contains:

Tilmicosin phosphate 250mg (potency)

Pharmacodynamics:

Tilmicosin is a semi-synthetic antibiotic of the macrolide group and is believed to affect protein synthesis. It has bacteriostatic action but at high concentrations it may be bactericidal. This antibacterial activity is predominantly against Gram-positive microorganism with activity against certain gram-negative ones and *Mycoplasma* of a bovine, porcine, ovine and avian origin. In particular, its activity has been demonstrated against the following microorganism:

- Pigs: *Mycoplasma hyopneumoniae*, *Pasteurella multocida* and *Actinobacillus pleuropneumoniae*
- Chicken and turkeys: *Mycoplasma gallisepticum* and *Mycoplasma synoviae*
- Calves: *Mannheimia haemolytica*, *Pasteurella multocida*, *Mycoplasma bovis* and *M. dispar*.

Scientific evidence suggests that macrolides act synergistically with the host immune system. Macrolides appear to enhance phagocyte killing of bacteria. Tilmicosin has been shown to inhibit in vitro the replication of the Porcine Reproductive and Respiratory Syndrome virus in alveolar macrophages in a dose dependent fashion.

Cross-resistance between tilmicosin and other macrolides and lincomycin has been observed.

Pharmacokinetics

Whilst blood concentrations of tilmicosin are low, there is pH-dependent macrophage accumulation of tilmicosin in inflamed tissues.

Pigs: After oral administration of 200mg tilmicosin/ litre drinking water, the average active substance concentrations detected in lung tissue, alveolar macrophages and bronchial epithelium 5 days after the start of treatment were 1.44µg/ml, 3.8µg/ml and 7.4µg/g respectively.

Poultry (chicken-broilers & pullets): As early as 6 hours after oral administration of 75mg tilmicosin/ litre drinking water, the average active substance concentrations detected in lung and alveolar tissue were 0.63µg/g and 0.30µg/g respectively. 48 hours after the start of treatment, the tilmicosin concentrations in lung and alveolar tissue were 2.3 µg/g and 3.29µg/g respectively.

Calves: As early as 6 hours after oral administration of 25mg tilmicosin/kg body weight/day in milk replacer, an average active substance concentration of 3.1µg/g was detected in lung tissue. 78 hours after the start of treatment, the tilmicosin concentration in lung tissue was 42.7µg/g. therapeutically effective concentrations of tilmicosin were measured up to 60 hours after treatment.

Turkeys: After oral administration of 75mg tilmicosin/litre drinking water, the average active substance concentrations detected in lung tissue, air sac tissue and plasma 5 days after the start of treatment were 1.89µg/ml, 3.71µg/ml and 0.02µg/g respectively. The highest mean tilmicosin concentration detected for lung tissue was 2.19µg/g at 6 days; for air sac tissue it was 4.18µg/g at 2 days and in the plasma it was 0.172µg/g at 3 days.

Indication(s):

Swine: For the treatment and prevention of respiratory disease in pig herds, associated with *Mycoplasma hyopneumoniae*, *Pasteurella multocida*, *Actinobacillus pleuropneumoniae* and other organisms susceptible to tilmicosin.

Poultry (chicken-broilers & pullets): For the treatment and prevention of respiratory disease in chicken flocks, associated with *Mycoplasma gallisepticum* and *M. synoviae*.

Turkeys: For the treatment and prevention of respiratory disease in turkey flocks, associated with *Mycoplasma gallisepticum* and *M. synoviae*.

Bovine (cattle and calves): For the treatment and prevention of bovine respiratory disease, associated with *Mannheimia haemolytica*, *Pasteurella multocida*, *Mycoplasma bovis*, *M. dispar* and other organisms susceptible to tilmicosin.

Target Species

Poultry (chicken-broilers & pullets), turkeys, swine, bovine (cattle and calves).

Recommended Dose

Swine:

15-20mg tilmicosin per kg bodyweight for 5 days, i.e. 6-8 ml of Pulcosin Oral Solution for 100 kg bodyweight corresponding to 80 ml of Pulcosin Oral Solution per 100 litres of drinking water for 5 days.

Poultry (chicken-broilers & pullets):

15-20mg tilmicosin per kg bodyweight for 3 days, i.e. 6-8 ml of Pulcosin Oral Solution for 100 kg bodyweight corresponding to 30ml of Pulcosin Oral Solution per 100 litre of drinking water for 3 days.

Turkeys:

10-27mg tilmicosin per kg bodyweight for 3 days, i.e. 4-11 ml of Pulcosin Oral Solution for 100 kg bodyweight corresponding to 30ml of Pulcosin Oral Solution per 100 litre of drinking water for 3 days.

Bovine (cattle and calves):

12.5mg tilmicosin per kg bodyweight twice a day for 3-5 consecutive days, i.e. 1ml of Pulcosin Oral Solution for 20kg bodyweight twice a day, for 3-5 days.

If signs of disease do not significantly improve within 3-5 days, the diagnosis should be re-evaluated and treatment changed.

To avoid under-dosing body weight should be determined as accurately as possible. The uptake of medicated water depends on the clinical condition of the animals. In order to obtain the correct dosage the concentration of the product has to be adjusted accordingly.

Mode of Administration

Orally via drinking water or milk replacer.

Contraindications:

- 1) Do not allow horses and other equines access to drinking water containing tilmicosin.
- 2) Do not use in case of hypersensitivity to tilmicosin or to any of the excipients.

Warning and Precautions

Special warnings for target species

Must be diluted in drinking water or milk replacer before administration. Protect from light after reconstitution. For oral use only. Tilmicosin should not be administered by injection to pigs.

Pigs, chickens and turkeys: Water consumption should be monitored in order to guarantee adequate dosing. In case water consumption does not match quantities for which recommended concentrations were calculated, the concentration of Pulcosin Oral Solution has to be adapted in a way that the recommended dosage will be taken up by the animals or different medication should be considered.

Special precaution for use in animals.

For oral use only. Do not inject.

Severely ill individuals tend to drink less and may need simultaneous treatment, preferably by parenteral medication.

Inappropriate use of the product may increase the prevalence of bacteria resistant to tilmicosin and may decrease the effectiveness of treatment with tilmicosin-related substances. The use of the product should be based on susceptibility tests.

Medicated drinking water or milk replacer should be prepared fresh every 24 hours.

Official, national and regional antimicrobial policies should be taken into account when the product is used.

Special precaution to be taken by the person administering the veterinary medicinal product to animals

- Tilmicosin may induce irritation. Macrolides, such as tilmicosin, may also cause hypersensitivity (allergy) following injection, inhalation, ingestion or contact with skin or eye. Hypersensitivity to tilmicosin may lead to cross reactions to other macrolides and vice versa. Allergic reactions to these substances may occasionally be serious and therefore direct contact should be avoided.
- To avoid exposure during preparation of the medicated drinking water, wear overalls, safety glasses, and impervious gloves. Do not eat, drink or smoke when handling this product. Wash hand after use.
- In the case of accidental ingestion, wash out mouth immediately with water and seek medical advice. In the event of accidental skin contact, wash thoroughly with soap and water. In case of accidental eye contact, flush the eyes with plenty of clean, running water.
- Do not handle the product if you are allergic to ingredients in the product.
- If you develop symptoms following exposure, such as skin rash, you should seek medical advice and show the physician this warning. Swelling of the face, lips and eyes or difficulty in breathing are more serious symptoms and require urgent medical attention.

Interaction with Other Medicaments

Cross resistance between tilmicosin and other macrolide antibiotics and Lincosamides has been observed. Tilmicosin may lessen the antibacterial activity of β -lactam antibiotics. Do not use simultaneously with bacteriostatic antimicrobial agents.

Pregnancy and Lactation

The safety of tilmicosin has not been established in animals used for breeding purposes. Use only in accordance with risk/benefit assessment by the responsible veterinarian. Do not use in pullets and laying hens producing egg for human consumption.

Side Effect(s):

In very rare cases, a decrease in water intake has been observed.

Environmental Property

None has been mentioned in animal.

Symptoms and Treatment for Overdosage, and Antidote(s):

Pigs drink less water when a dose of 300 to 400mg per litre (1.5 to 2 times the recommended dose) is administered. Although this will result in less intake of tilmicosin, it might lead to dehydration of the animals. Replace with untreated water when needed.

No symptoms were seen in poultry treated at 375mg per litre during 5 days. A dose of 75mg per litre during 10 days resulted in less consistent faeces.

No symptoms of overdose were noticed in turkeys treated at 375mg per litre of drinking water during 3 days. No symptoms were noticed at 75mg per litre during 6 days.

Except for a slight decrease in milk intake, no symptoms of overdose were seen in calves treated at 5 times the recommended dose or during twice the recommended treatment period.

Storage Condition:

Store at temperature below 30°C. Protect from direct sunlight and water.

Shelf Life:

2 years.

After reconstituted with water, the medicated solution must be used within 24 hours.

Withdrawal Periods

Meat and offal:

Swine: 14 days

Bovine: 42 days

Chicken (broilers): 12 days

Turkeys: 19 days

Eggs: Do not administer to poultry producing eggs for human consumption. Do not use within 14 days of onset of the laying.

Not for use in lactating animals.

Maximum Residual Limit (MRL)

Muscle, fat (cattle, poultry and pig) 100 µg/kg

Kidney (cattle) 300 µg/kg

Liver (cattle) 1000 µg/kg

Liver (pig) 1500 µg/kg

Kidney (pig) 1000 µg/kg

Product Description and Packing(s):

A dark brownish clear solution.

Plastic bottle of 1L and 5L.

**Made by YSP**

Manufacturer:

Y.S.P INDUSTRIES VIETNAM CO., LTD

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Product Registration Holder :

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